



Checking and securing understanding of non-linear sequences

1 A sequence with a multiplicative, term-to-term relationship is called _____ sequence. Tick **1** correct answer

☐ an arithmetic

☒ a geometric

☐ a linear

☐ a multiplier

2 Match the sequence to the descriptor. Write the correct letter in each box

a	4, 8, 12, 16, ...
b	2, 4, 8, 16, ...
c	1, 4, 8, 13, ...

a	Arithmetic sequence
b	Geometric sequence
c	Other sequence

3 Find the common ratio of the geometric sequence 192, 48, 12, ... Tick **1** correct answer

☐ 4

☐ 0.4

☒ $\frac{1}{4}$

☐ 2

☐ $\frac{1}{2}$



- 4 What is the common second difference of the sequence 3, 12, 27, 48, ... ? Tick **1** correct answer

☐ 4☒ 6☐ 9☐ 15

- 5 Select the statements that are true of the sequence 2, 16, 54, 128, 250, ... Tick **1** correct answer

☐ It is geometric. It has a common ratio.☐ It is arithmetic. It has a common first difference between terms.☐ It has a common second difference.☒ It has a common third difference.☐ It has a common fourth difference.

- 6 Which of these sequences will not always decrease? Tick **1** correct answer

☐ 1240, 620, 310, 155, ...☐ 62, 51, 40, 29, ...☒ 104, 90, 80, 74, ...